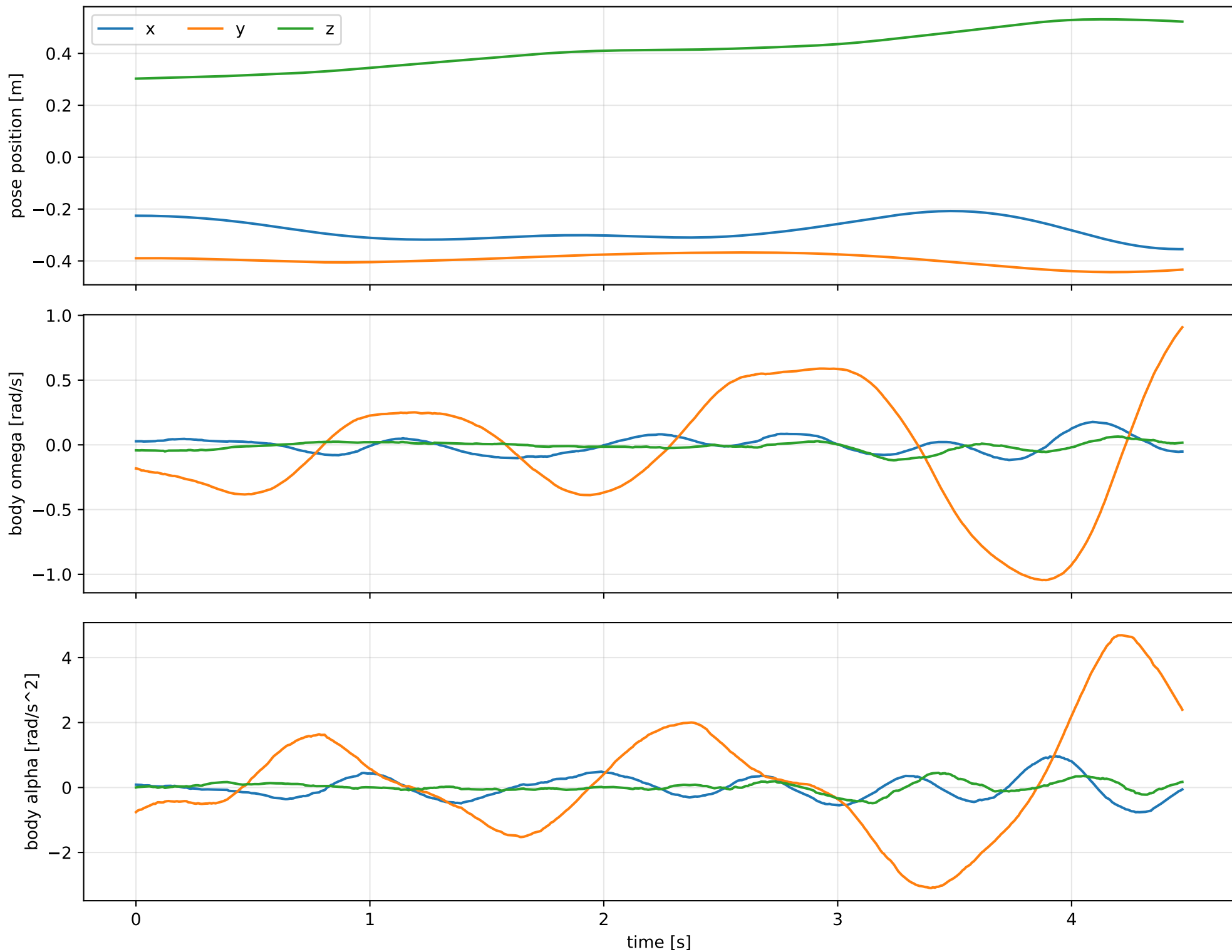
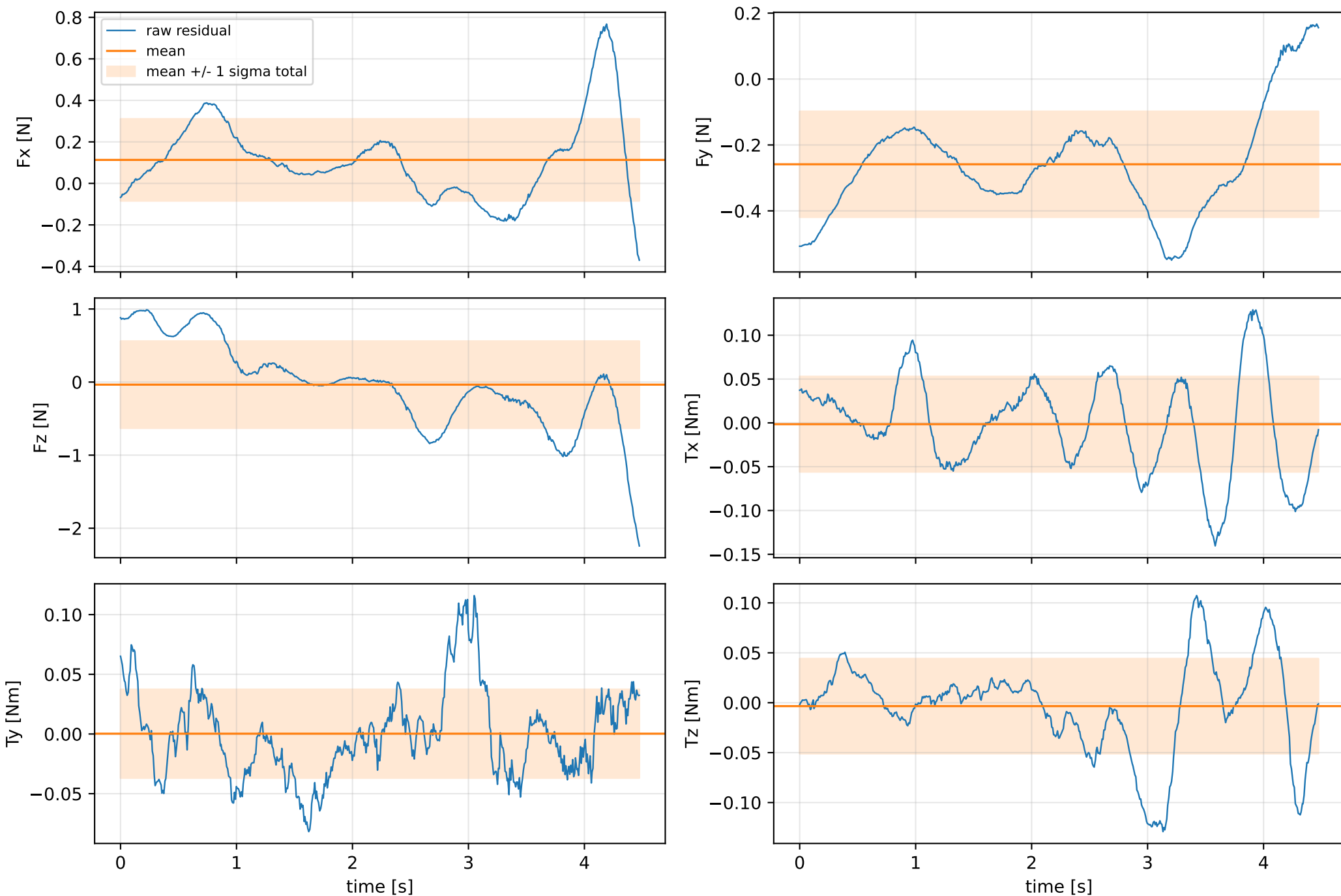


inertia_over_mass_xz_nominal: SG trajectory and derived motion



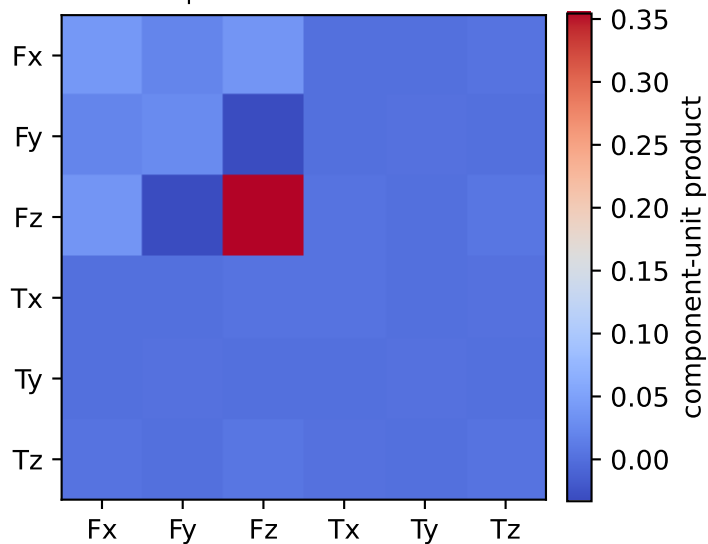
inertia_over_mass_xz_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



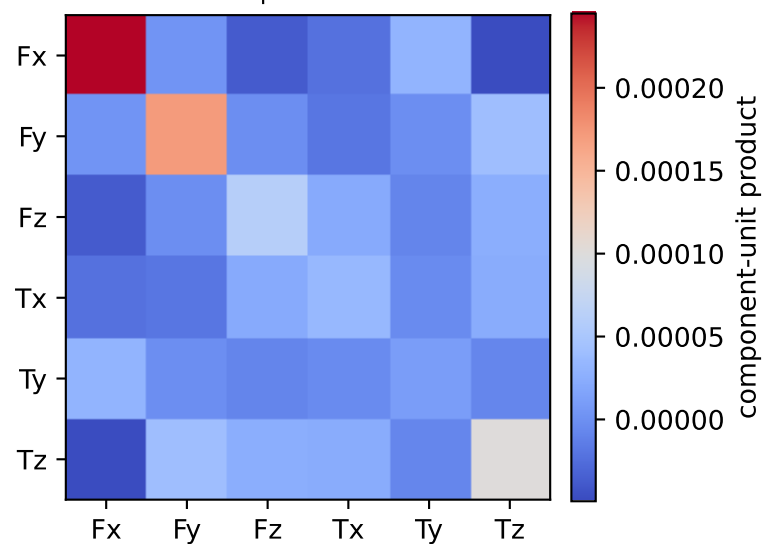
vector RMS about zero: force 0.70656 N, torque 0.08122 Nm; component std about mean: force [0.1974 0.1605 0.5953], torque [0.0544 0.0371 0.0475]

inertia_over_mass_xz_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.004 0.005 0.052 0.361]

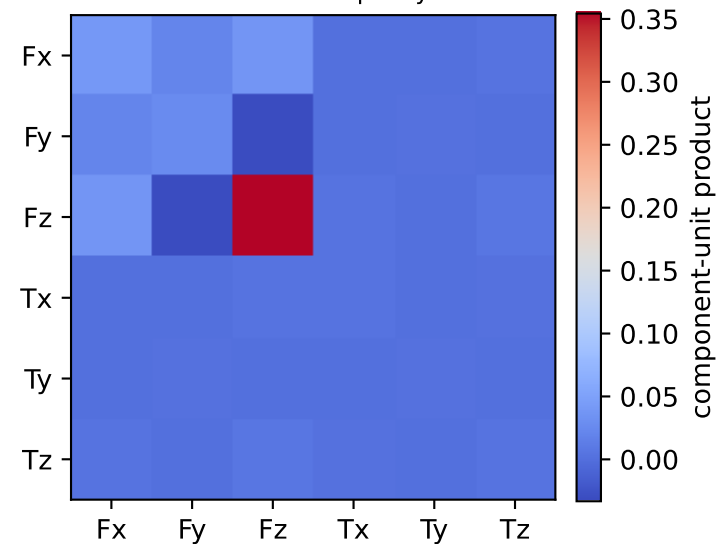
empirical total covariance



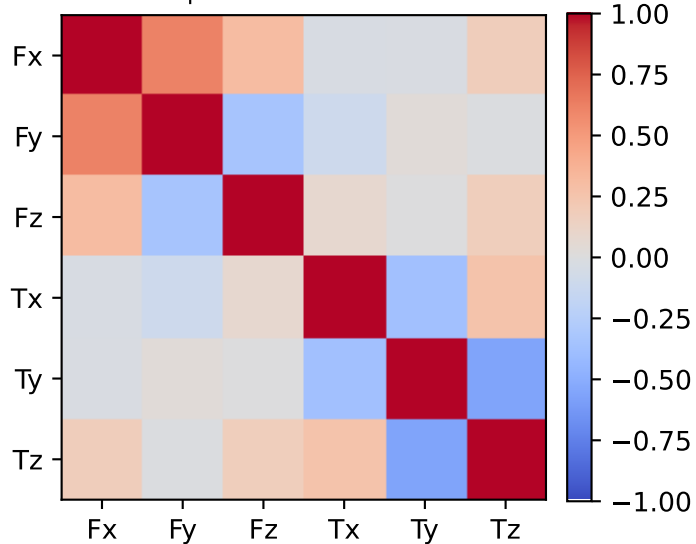
mean full-SG prediction covariance



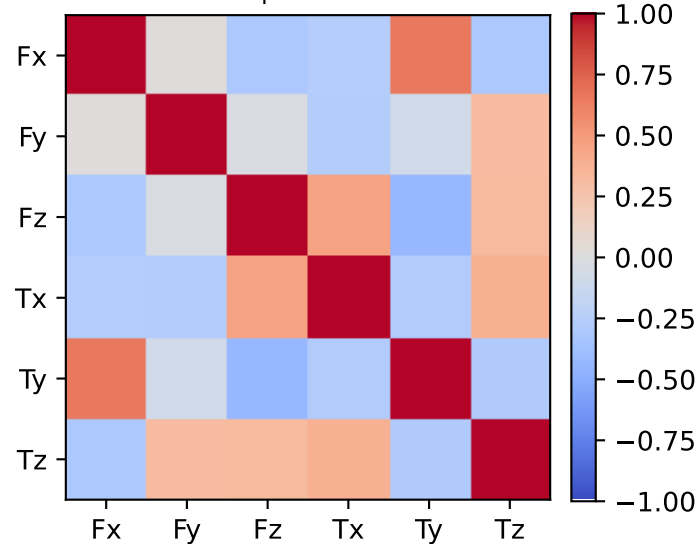
PSD excess model discrepancy covariance



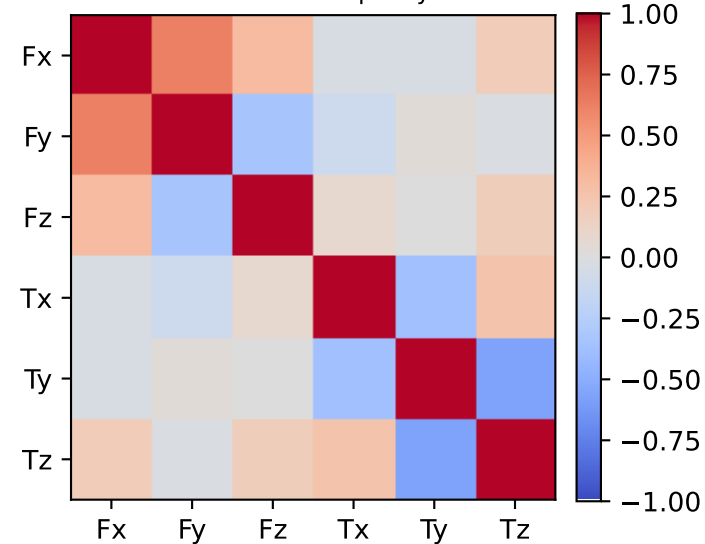
empirical total correlation



mean full-SG prediction correlation

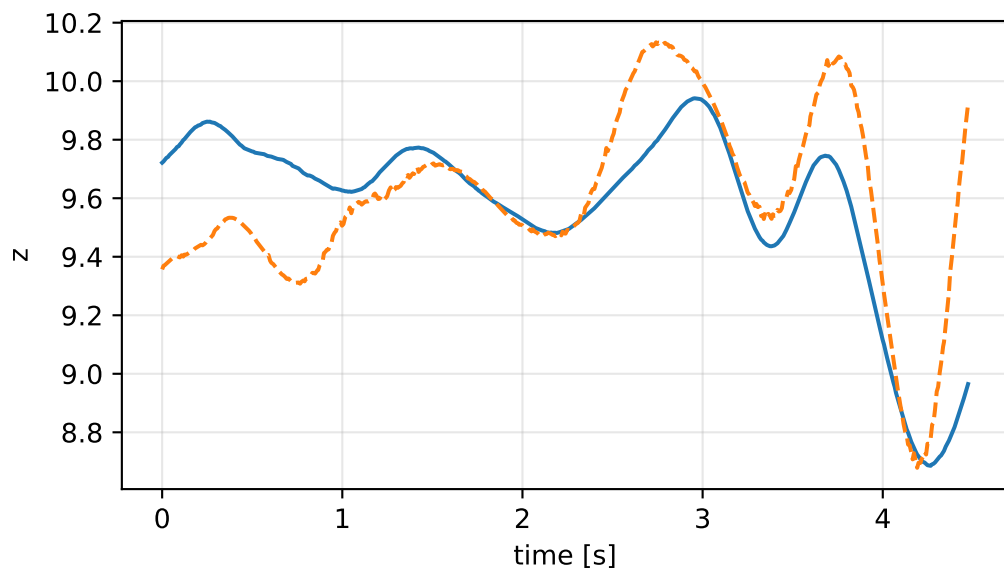
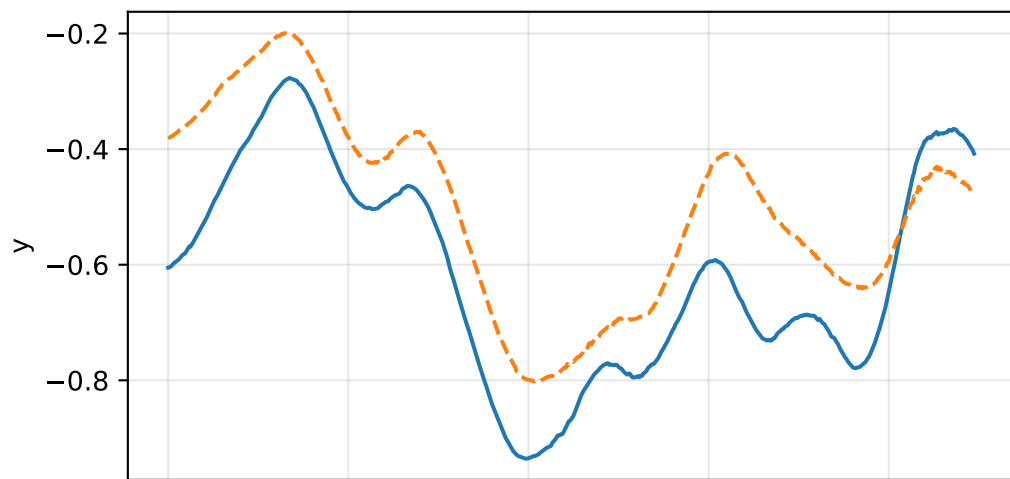
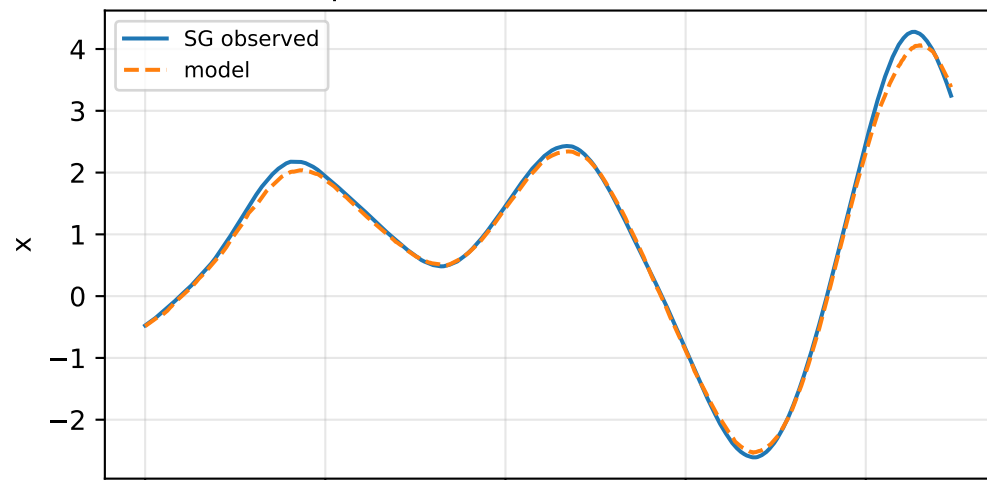


PSD excess model discrepancy correlation

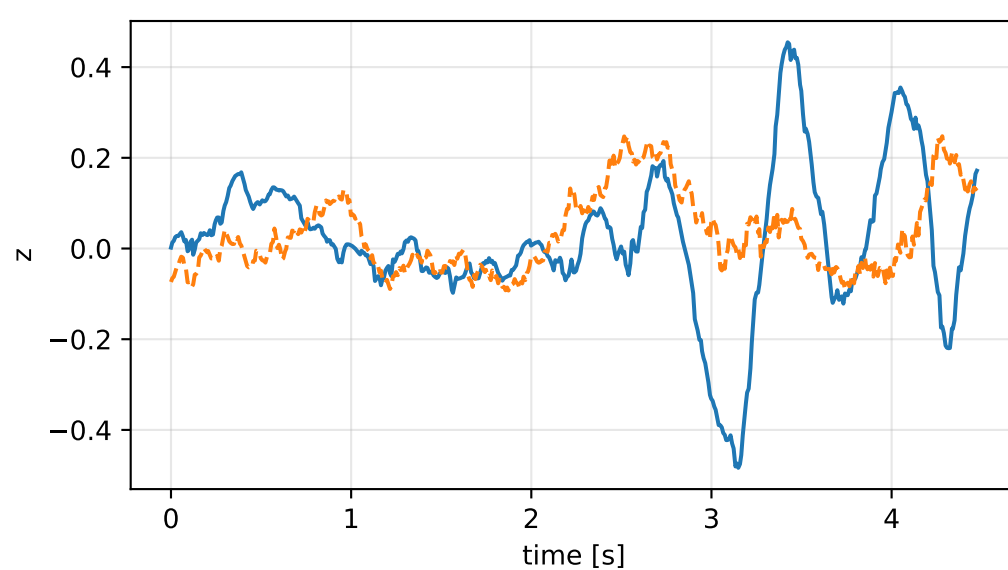
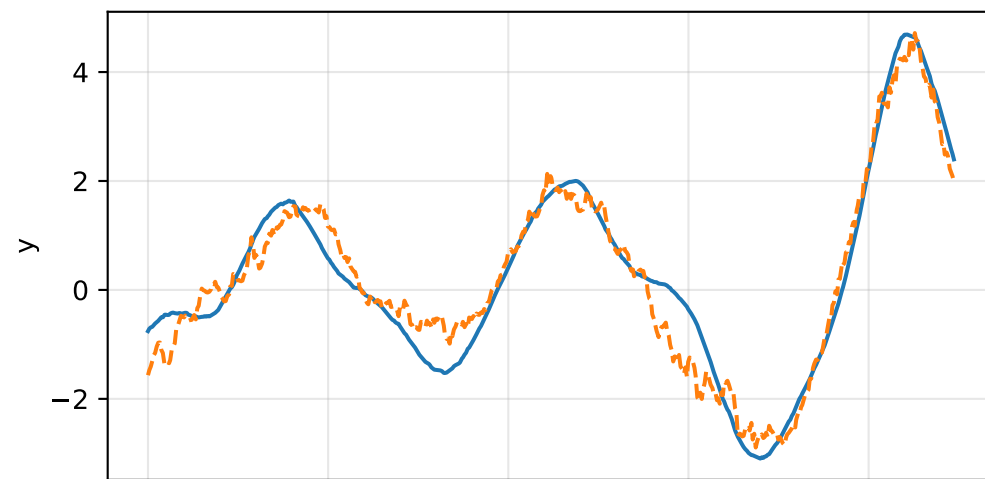
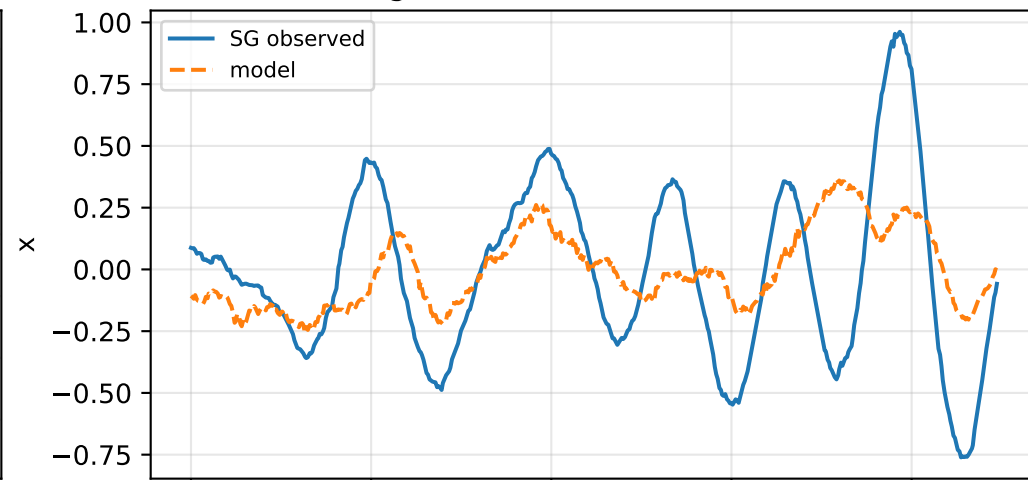


inertia_over_mass_xz_nominal: acceleration residual objective

specific acceleration [m/s^2]

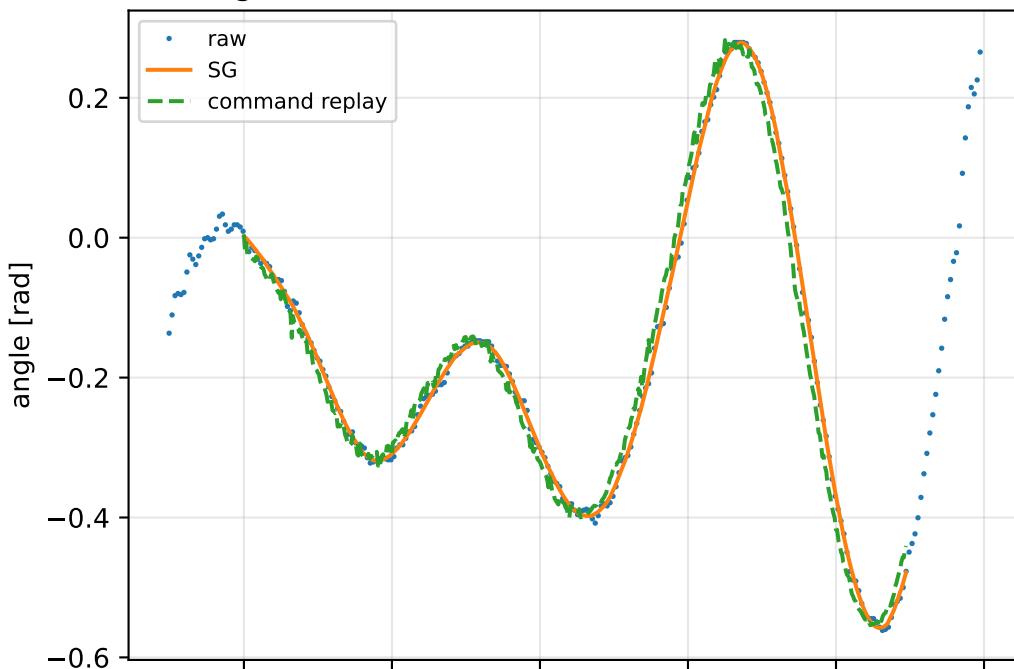


angular acceleration [rad/s^2]

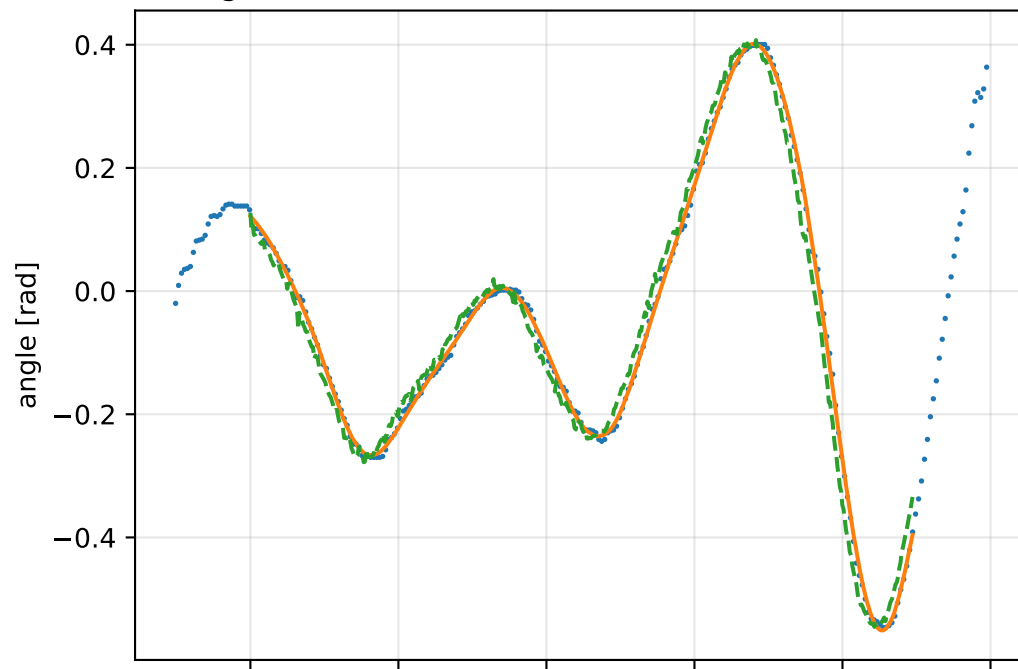


inertia_over_mass_xz_nominal: gimbal measurement audit (objective=measured_sg)

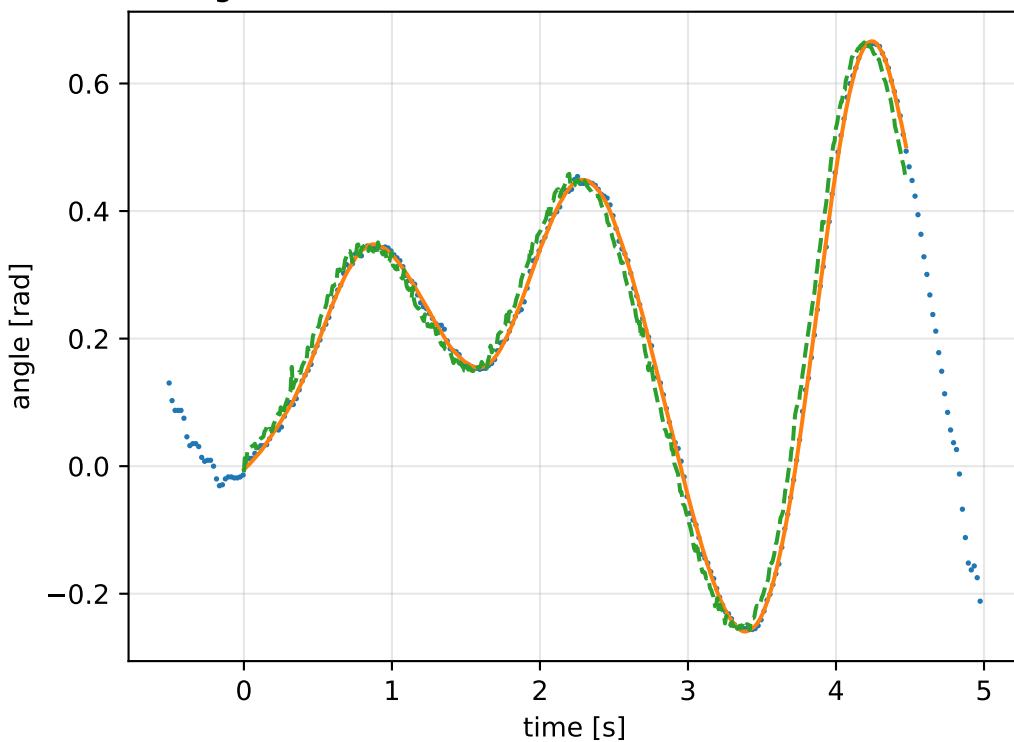
gimbal 1: RMSE=0.027, max=0.07529 rad



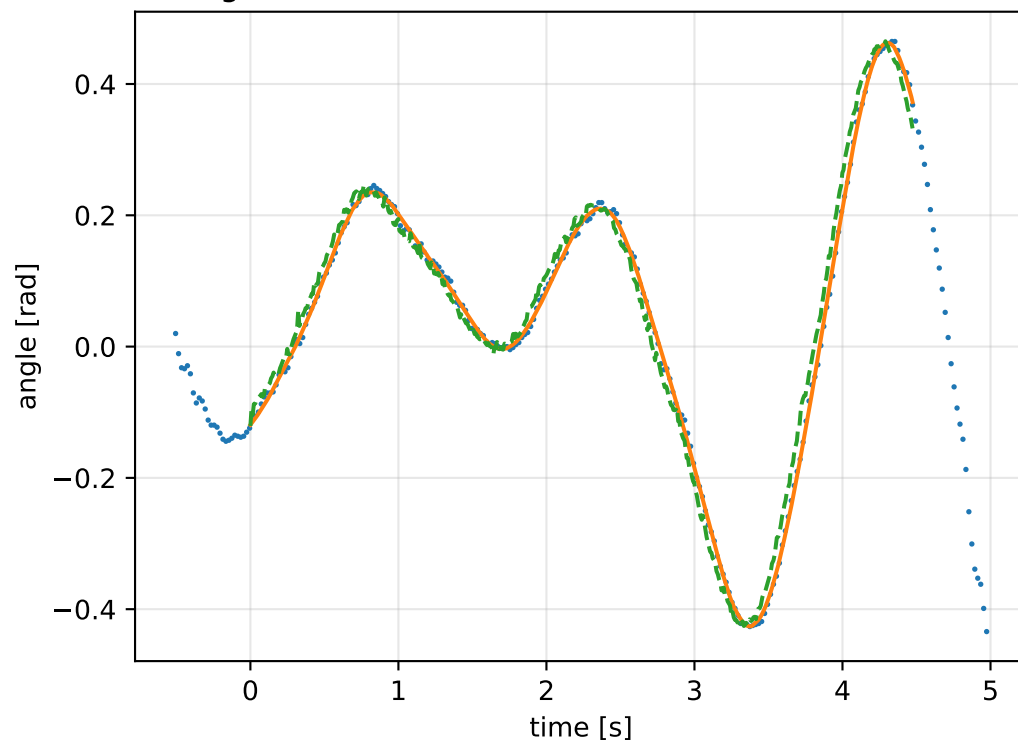
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

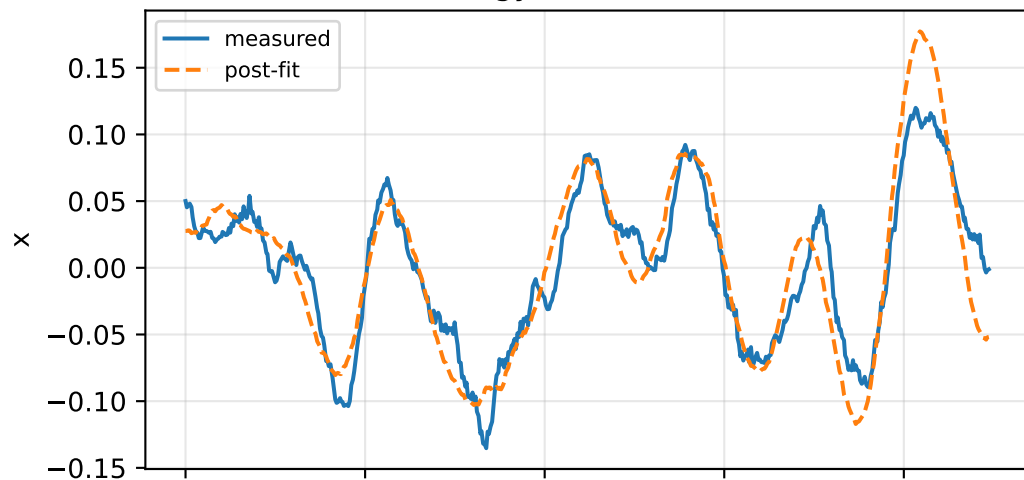


gimbal 4: RMSE=0.02598, max=0.07261 rad

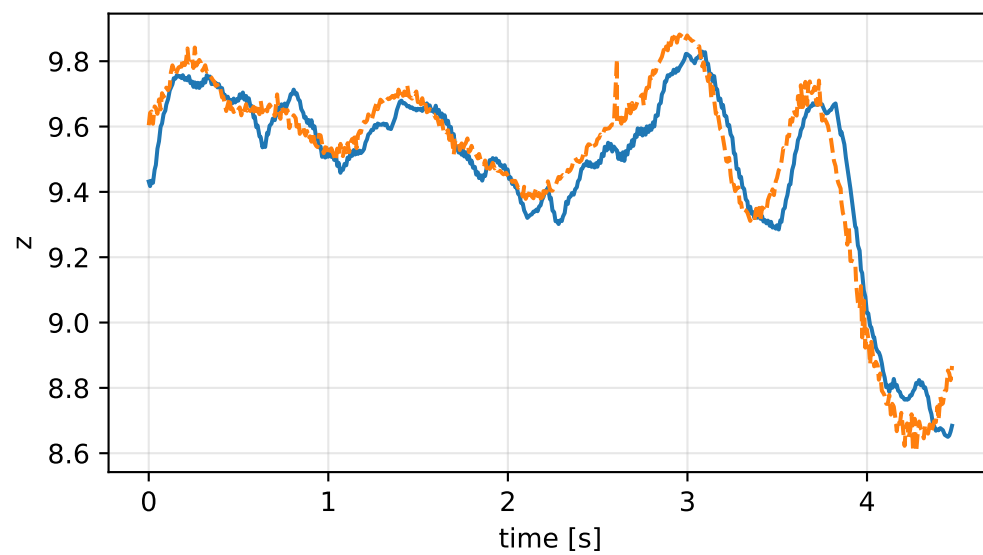
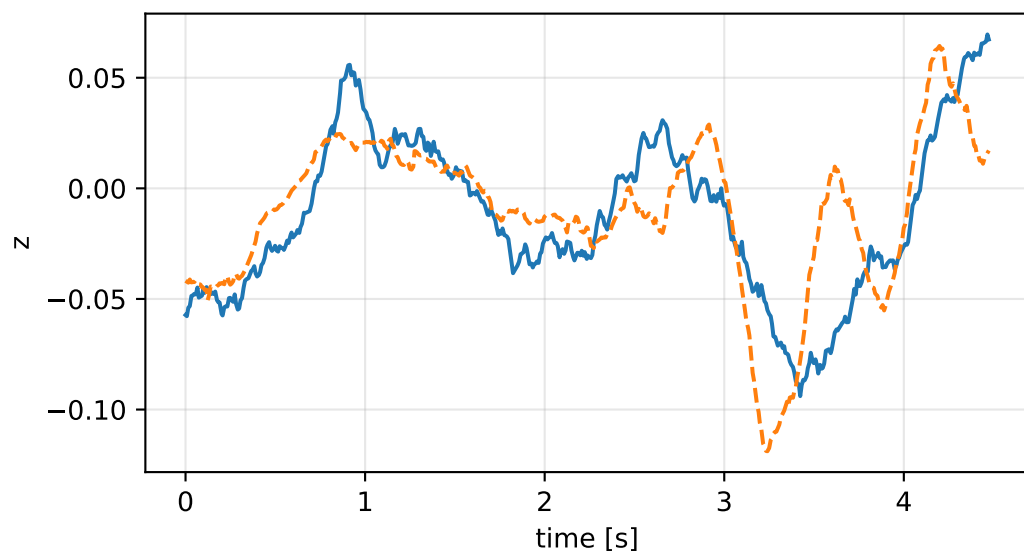
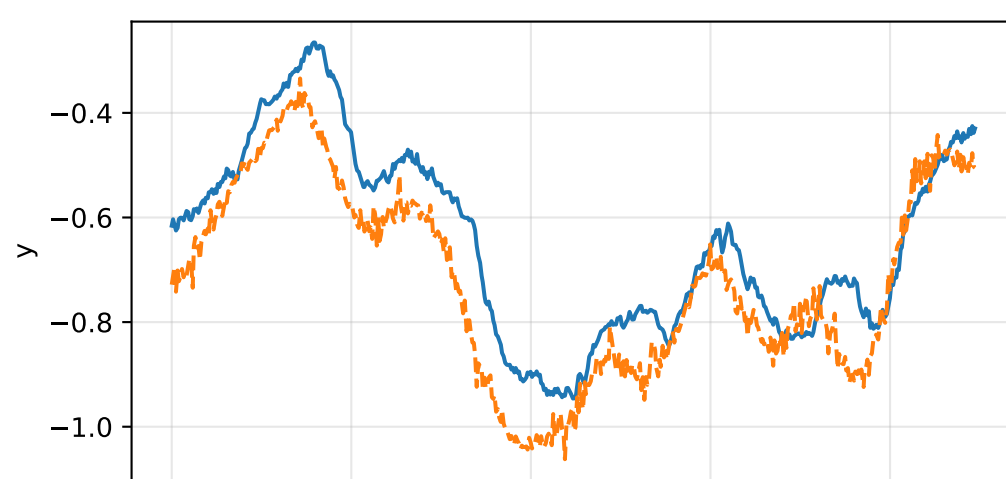
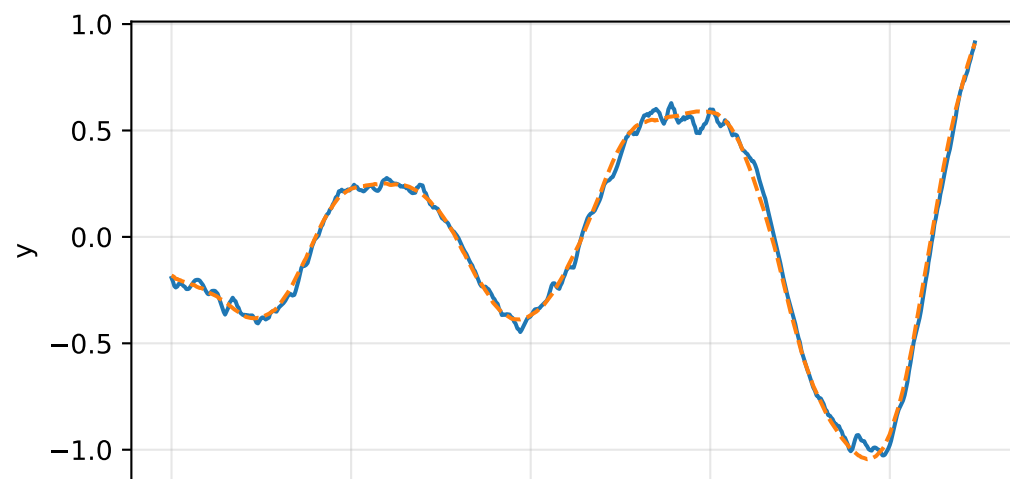
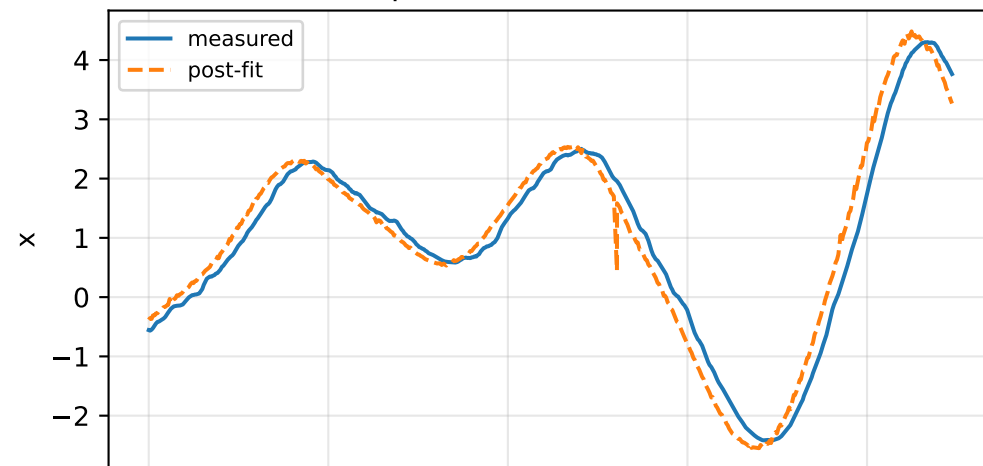


inertia_over_mass_xz_nominal: IMU comparison (diagnostic only)

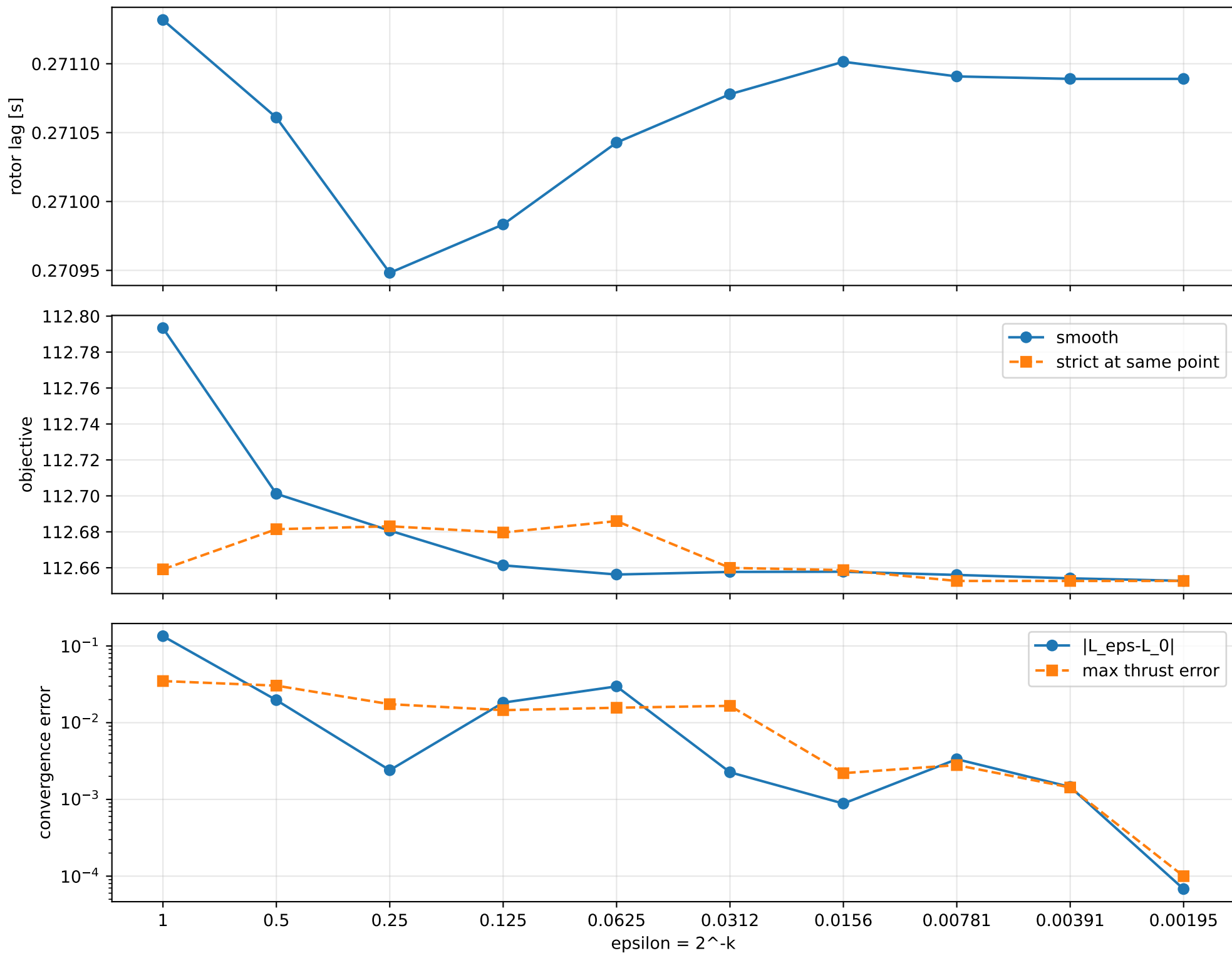
gyro [rad/s]



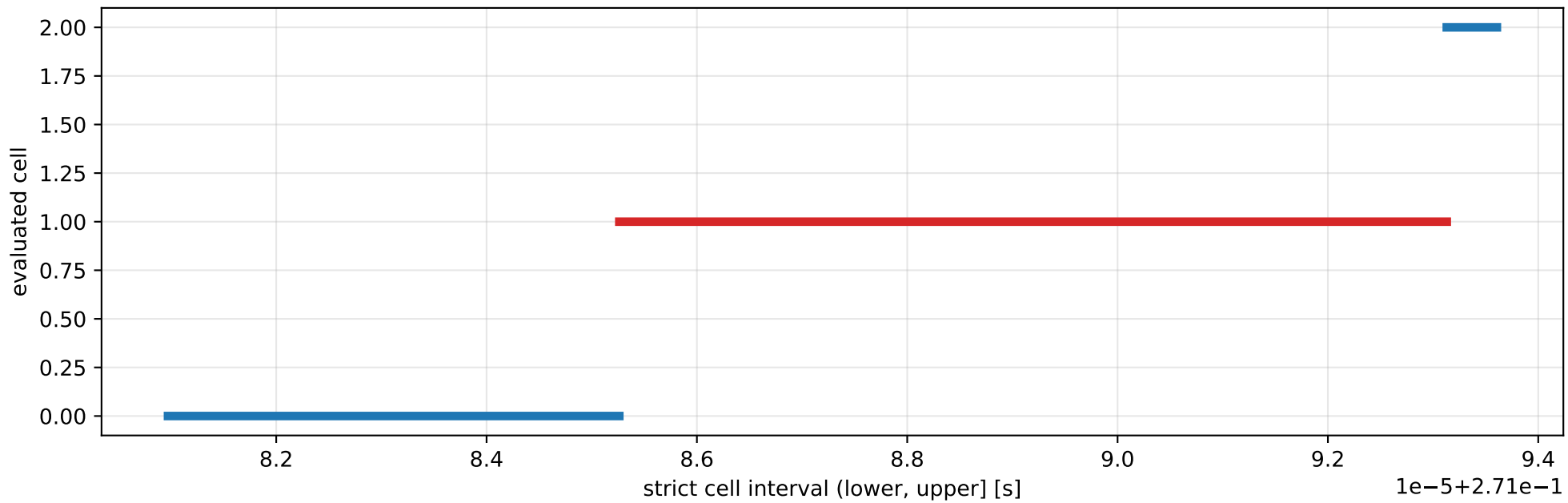
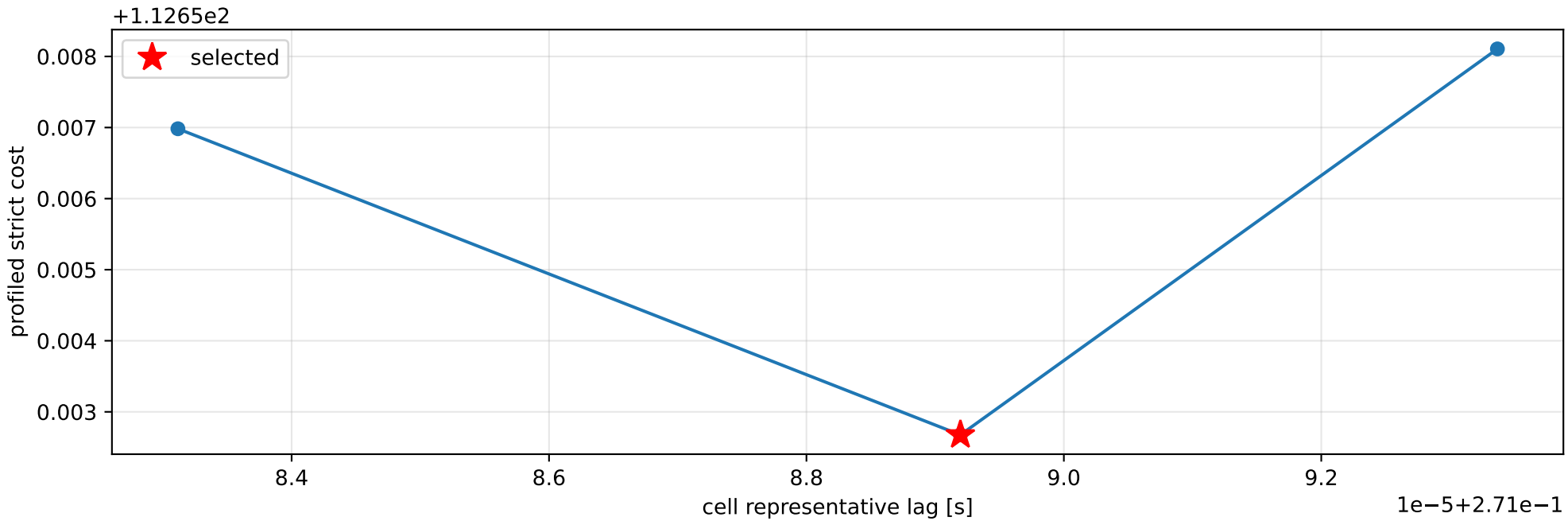
specific force [m/s²]



inertia_over_mass_xz_nominal: smooth-to-strict continuation



inertia_over_mass_xz_nominal: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089013 s; data boundary=False

inertia_over_mass_xz_nominal: parameter estimate

Case: inertia_over_mass_xz_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.48040717

inertia [kg m²]:

[[4.04749150e-01 9.93007458e-03 4.43146903e-05]

[9.93007458e-03 1.90812585e-01 -5.99875551e-02]

[4.43146903e-05 -5.99875551e-02 5.77345833e-01]]

CoG body [m]: [-0.02646295 0.0114242 0.00939665]

force effectiveness: [2.09223063 1.69215234 1.86888449 2.24462774]

scale-free J/m [m²]:

[[7.38538466e-02 1.81192278e-03 8.08602153e-06]

[1.81192278e-03 3.48172279e-02 -1.09458209e-02]

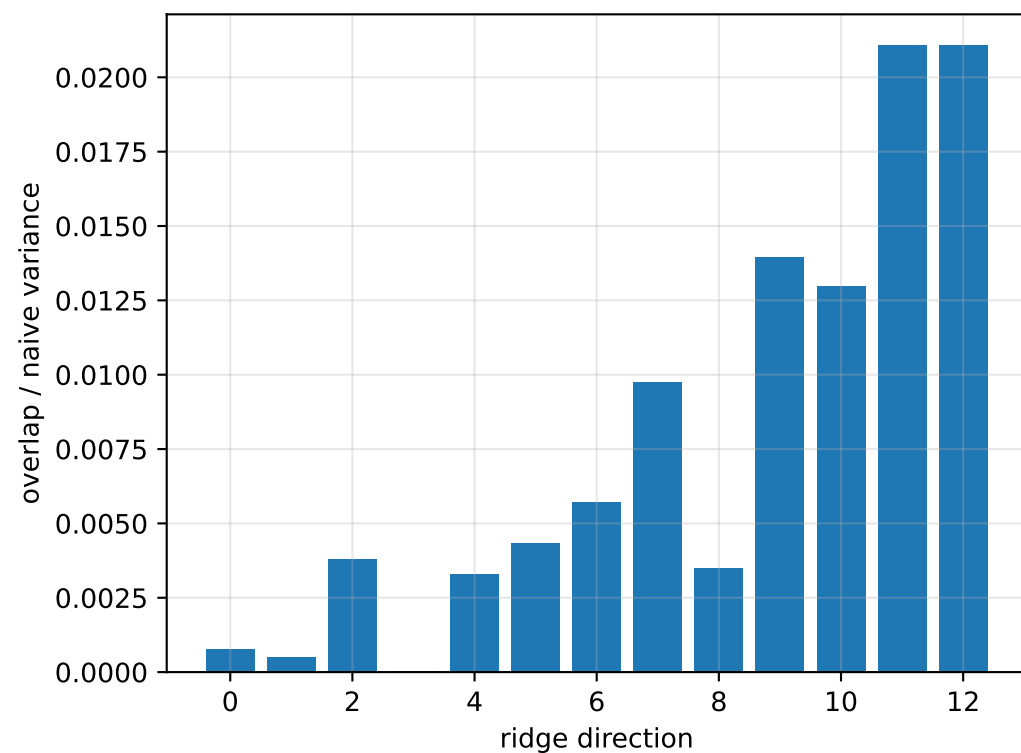
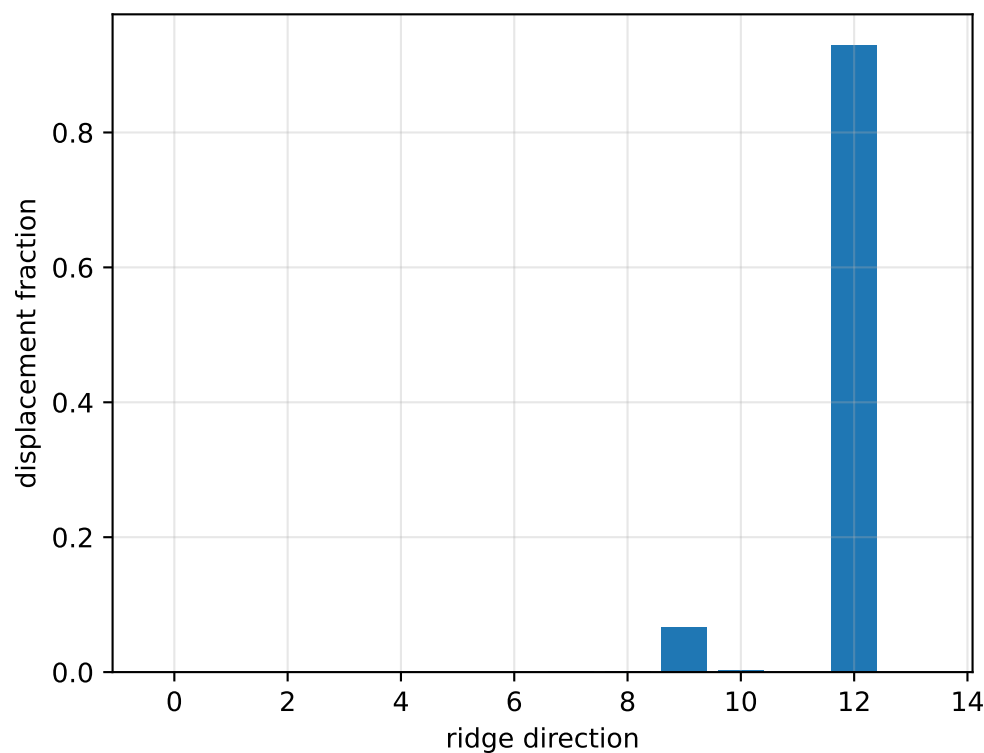
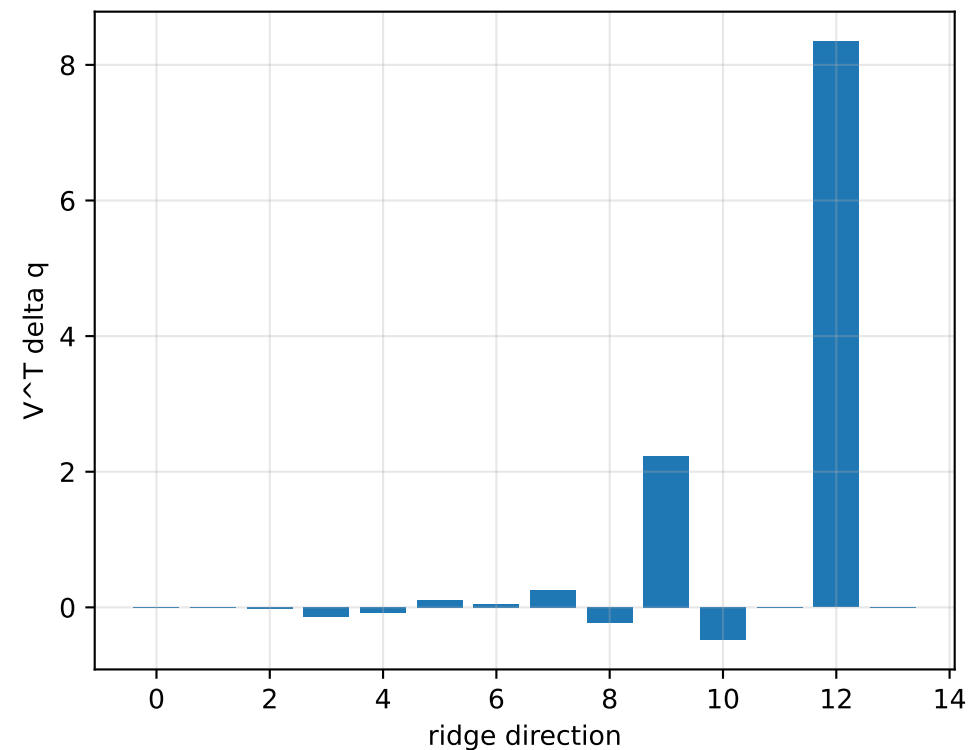
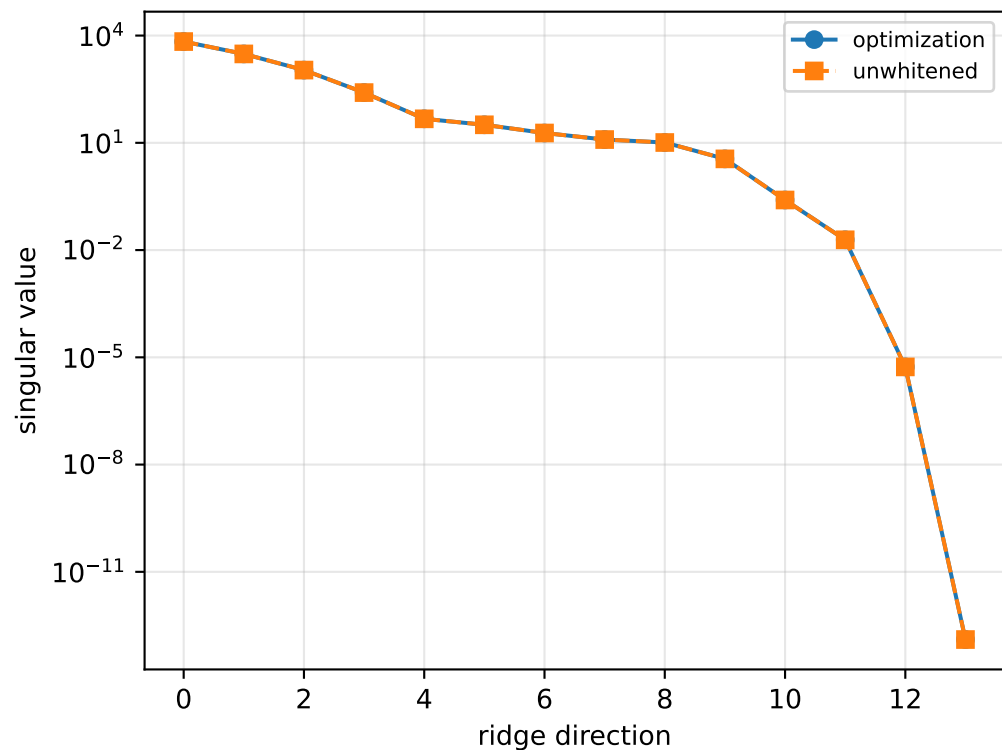
[8.08602153e-06 -1.09458209e-02 1.05347252e-01]]

scale-free f/m: [0.38176554 0.30876398 0.34101198 0.40957317]

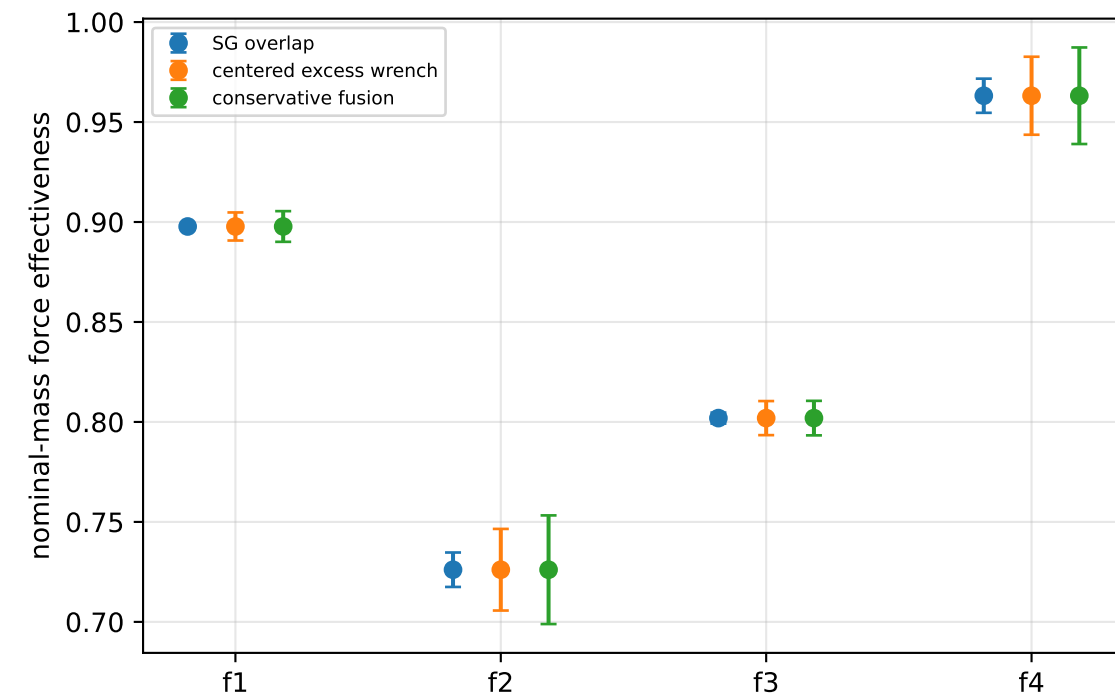
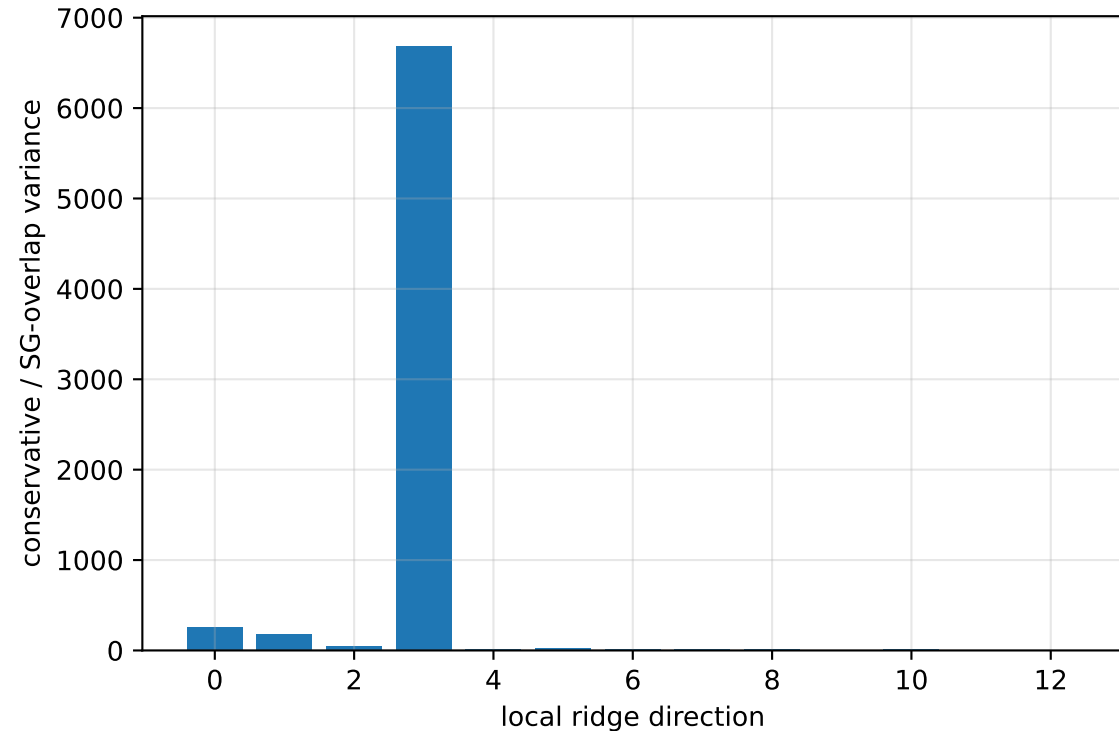
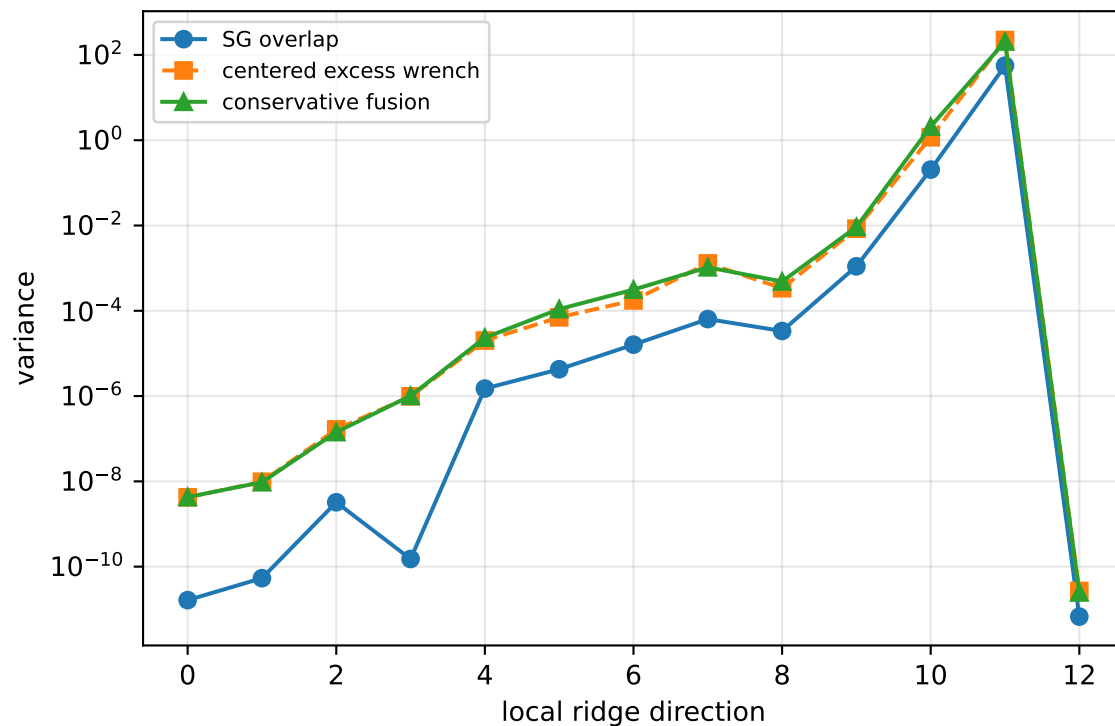
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

inertia_over_mass_xz_nominal: ridge spectrum (rank 13, nullity 1, || $\mathbf{v}$ ||=1.41e-13)



inertia_over_mass_xz_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 229.4$
wrench-to-acceleration recovery max error = $2.83\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

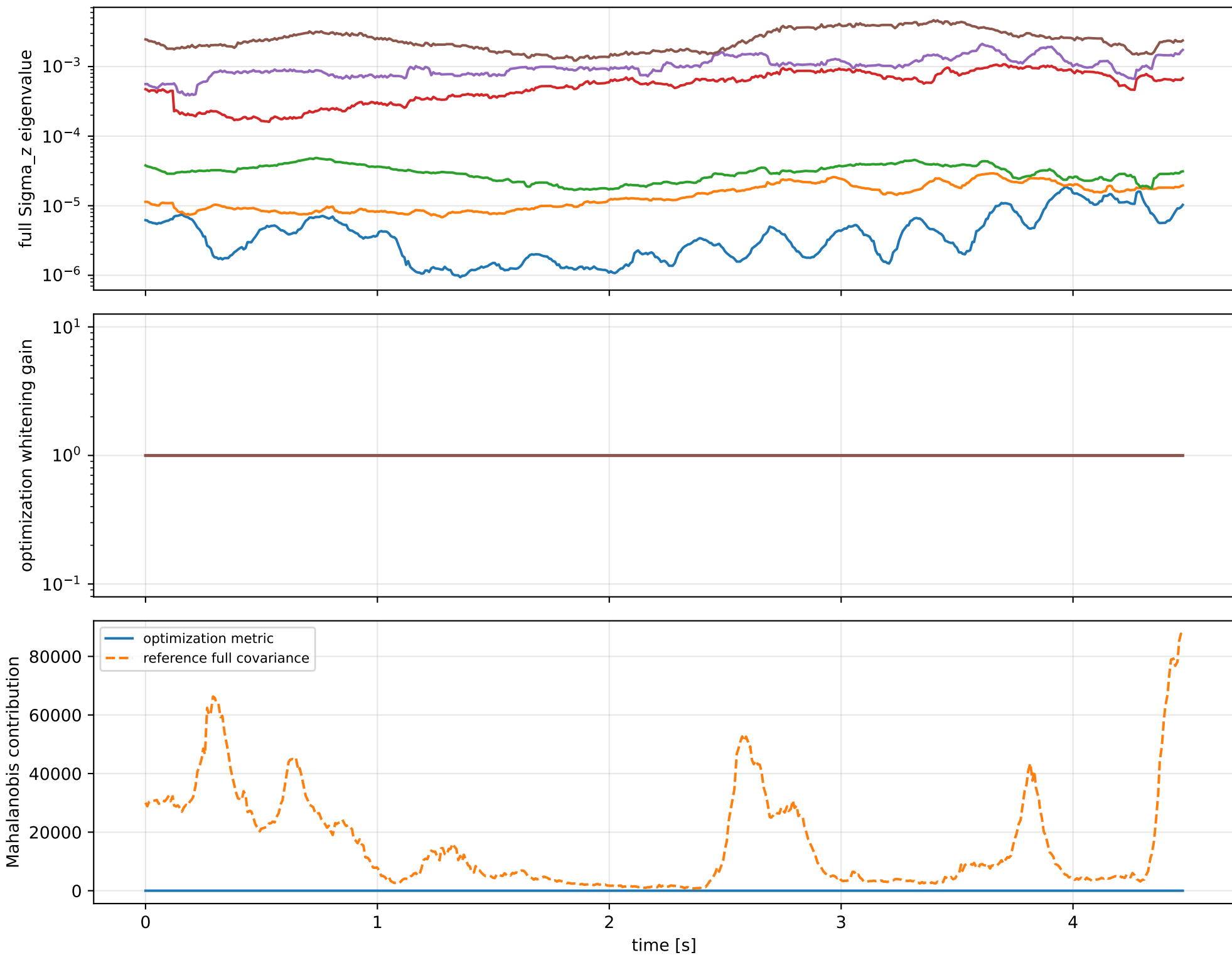
index 11: variance=206.3, ratio=3.685
 $\log_{\text{mass}}=+0.00112$; second_moment_diag_1=+0.0602
second_moment_diag_2=-0.0427; second_moment_diag_3=-0.023
second_moment_offdiag_12=+0.838; second_moment_offdiag_13=-0.539
second_moment_offdiag_23=+0.048; cog_x=-6.56e-05
cog_y=+3.89e-05; cog_z=-9.18e-06
log_force_eff_1=+0.00122; log_force_eff_2=+0.000571
log_force_eff_3=+0.000988; log_force_eff_4=+0.00153

index 10: variance=2.092, ratio=10.22
 $\log_{\text{mass}}=+0.0303$; second_moment_diag_1=+0.0333
second_moment_diag_2=+0.548; second_moment_diag_3=-0.738
second_moment_offdiag_12=+0.00669; second_moment_offdiag_13=+0.0366
second_moment_offdiag_23=+0.384; cog_x=+0.00117
cog_y=-0.00161; cog_z=+0.000361
log_force_eff_1=+0.0312; log_force_eff_2=+0.0461
log_force_eff_3=+0.0274; log_force_eff_4=+0.0209

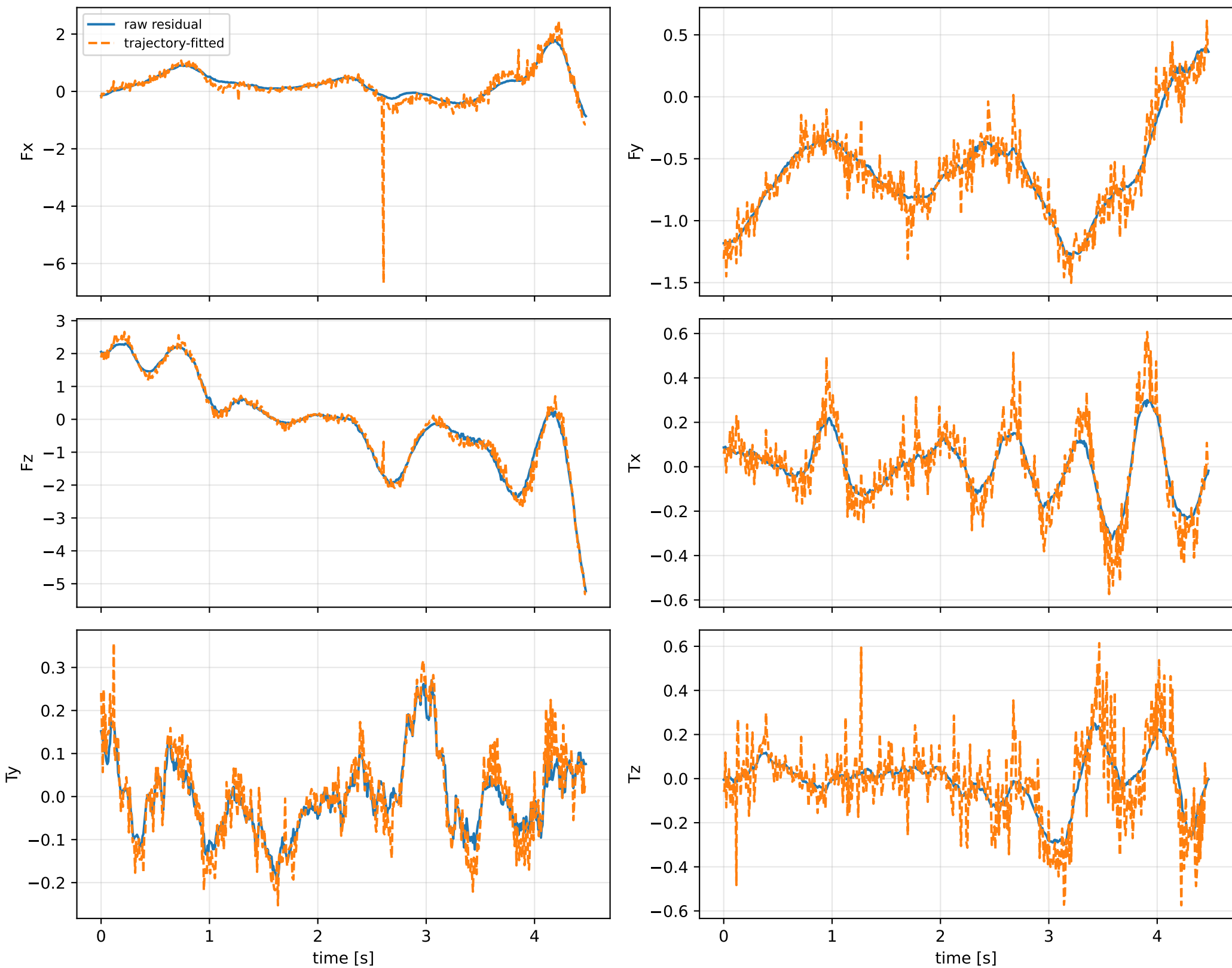
index 9: variance=0.009234, ratio=8.377
 $\log_{\text{mass}}=-0.139$; second_moment_diag_1=-0.134
second_moment_diag_2=+0.341; second_moment_diag_3=+0.545
second_moment_offdiag_12=-0.00791; second_moment_offdiag_13=-0.0215
second_moment_offdiag_23=+0.643; cog_x=-0.0199
cog_y=+0.0106; cog_z=-0.00279
log_force_eff_1=-0.102; log_force_eff_2=-0.299
log_force_eff_3=-0.187; log_force_eff_4=-0.0231

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_xz_nominal: optimization vs reference covariance



inertia_over_mass_xz_nominal: wrench / closure (force max 1.55e-14, torque max 2.08e-16)



inertia_over_mass_xz_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_xz_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: 99d969ed0e0fa85ae900b089864493fcb7e345fdd5fd63009a6ebdbc6ee04171

data objective: 112.652675944

prior objective: 1.15272243442e-11

total objective: 112.652675944

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_xz_nominal (inertia_over_mass_m2)

components: xz

target: [8.08602633e-06]

std: [1.e-06]

final: [8.08602153e-06]

error: [-4.80150484e-12]

standardized residual: [-4.80150484e-06]

factor objective: 1.15272243442e-11